OligoTox-Coagulopathy — Methodology

Materials and methods · NIH/NCATS Oligonucleotide Toxicity Open Data Challenge, Phase 2

Materials and methods for the coagulation-toxicity dataset, NIH/NCATS Oligonucleotide Toxicity Open Data Challenge, Phase 2.

1. Design

This is a **curated dataset**. No wet-lab experiment was performed. The methods below are therefore of two kinds, kept strictly apart: **(a)** the experimental methods of the source studies, recorded as each source states them, and **(b)** the curation methods used to find, extract, verify and harmonise them. Conflating the two would let curation choices masquerade as experimental fact.

Snapshot: **213 oligonucleotides · 2,388 measurements · 941 per-position modification records · 75 sources.**

2. Scope — what counts as this endpoint

Included: clotting times (aPTT, PT/INR, TT, ACT), fibrinogen, D-dimer, thrombin generation, coagulation-factor and antithrombin activity, anti-Xa/anti-IIa activity, and bleeding or thrombotic outcomes a source attributes to a coagulation defect.

Excluded, deliberately: **platelet count alone**. Thrombocytopenia is a separate Challenge endpoint with its own dossier at `../thrombocytopenia.md`. A platelet-related row appears here only where the source ties it to a cascade readout, and says so in `notes`. This boundary was set before extraction, after finding that the repository's largest existing store of "coagulation-adjacent" text — 16 hits on one page of a review already held — is entirely thrombocytopenia and belongs to that endpoint.

Both **unintended coagulopathy** and **on-target anticoagulant/procoagulant pharmacology** are in scope, on separate flags, because the second is where nearly all published per-compound clotting numbers live and because bleeding risk from an on-target anticoagulant is a genuine safety signal. They are never pooled: see README.

3. Source identification

Eight independent search axes were run in parallel — mechanism/in-vitro assays; clinical trials; nonclinical toxicology; siRNA and LNP; aptamers; regulatory labels; patents; reviews used as a citation map. Each axis was required to **retrieve and open** a candidate and quote the coagulation passage before reporting it; nothing entered the work-list on the strength of a title or a search summary. 96 raw candidates were merged to 73 distinct sources, then triaged into an extraction work-list ranked by whether the source gives per-compound numbers, per-compound qualitative statements, or class-level prose only.

Two results of that pass are worth recording because they are negative:

- **The project's only previously recorded lead was closed.** `coagulopathy.md` named Crooke et al. (2016), *Mol Ther* 24(10):1771–1782, as the single lead for this endpoint, on the strength of its title. It was retrieved and read: it does not report the per-compound coagulation values the title implies. It is recorded as context, not as a source of rows.
- Three sources whose granularity two axes disagreed about were resolved **downward**, to qualitative, because their values live only in figure panels.

4. Retrieval

All documents were retrieved directly; no value in this dataset comes from a search-engine summary. This is a deliberate break from the sibling kidney dataset, which was built under a network policy that blocked outbound fetch and whose own validation file concluded that its search-derived clinical rows had produced a provenance/outcome confound. Routes used:

Route	Sources
PMC / Europe PMC open-access JATS full text	27
NLM DailyMed SPL XML (US prescribing information)	8
NCBI eutils PubMed abstract (where no open full text exists)	8
USPTO grant text	7
Other staged full text and supplementary files	25

Every retrieved document is committed to `sources/documents/`, so each row's evidence can be re-read from the repository without network access. One supplementary PDF could not be retrieved; the 14 rows citing it are flagged and excluded from source verification rather than quietly passed.

5. Extraction

Sources were divided into 14 bundles by compound family and document type, and read individually. Extraction rules, enforced by the output contract:

- Every measurement carries a **verbatim quote** and an **exact locus** (table number, figure panel, section heading, label section, page). A row that could not be quoted was not written.
- **No value was read off a figure.** Where a number exists only in a plotted panel, the row carries `NOT_REPORTED` with `readout_is_qualitative = TRUE`. This cost real yield — one source's entire PT/aPTT time course is a figure — and the loss is recorded rather than recovered by pixel-reading.
- **Per-position chemistry is transcribed, never modelled.** A `modifications` row exists only where the source publishes position-resolved chemistry, and each row records in `basis` how the position was determined, normally the source's own case legend quoted verbatim.
- Severity is recorded **in the source's own words**; no grade was assigned during

extraction.

6. Oligonucleotide identity, purity and characterisation

The Challenge requires the methods used to purify and characterise oligo identity. The compounds were synthesised by the source laboratories, so what can be reported is what each source states, and `purity_pct` is **`NOT_REPORTED` for all 213 compounds**. No purity value was estimated, inferred from a synthesis platform, or carried across from another compound. Where a source states a synthesis platform, purification method or identity confirmation, it populates `synthesis_platform`, `purity_method` and `identity_confirmation`. This absence is a property of the published literature, not of the curation: per-compound purity is almost never published alongside toxicity results. Identity here means the printed sequence together with its per-position chemistry. 97 of 213 compounds have a published sequence; 47 have position-resolved chemistry. QC checks that every declared length equals the actual string (plus any documented terminal residue) and that every modification row's nucleobase matches the sequence at that position.

7. Harmonisation and grading

Categorical fields use the controlled vocabularies in `schema.md`. Grading is mechanical, from the control-referenced ratio, by **CTCAE v5.0** cut-offs — a published standard, not thresholds invented here — and only for the readouts CTCAE defines. 942 of 2,388 rows are graded; the remaining 1,446 each state in `grade_basis` why no published rule applies. The one deviation, applying CTCAE's limit-of-normal ratios to a matched-control ratio, is named in every graded row's `grade_basis` so it cannot be overlooked. All grades are `provisional`.

8. Predictor variables and their distribution

Variable	Distribution (n = 213 compounds)
Class	aptamer 59 · ASO gapmer 58 · other 36 · GalNAc-siRNA 28 · siRNA 13 · tcDNA-ASO 7 · PMO 4 · polydisperse ssDNA 3 · ASO mixmer 2 · CpG ODN 2 · SSO 1
Backbone	not reported 98 · full-PS 57 · mixed PO/PS 34 · full-PO 14 · PMO-neutral 4 · other/NA 4
Sequence published	97 / 213
Position-resolved chemistry	47 / 213 (941 position records)

9. Indicator variables and their distribution

Variable	Distribution (n = 2,388 measurements)
Readout category	clotting time 1,160 · factor activity 387 · bleeding outcome 240 · thrombotic outcome 220 · fibrinogen 144 · platelet–coagulation crosstalk 83 · anticoagulant activity 76 · thrombin generation 47 · fibrinolysis marker 31
Readout (top)	aPTT 599 · PT 376 · fibrinogen 140 · FXI activity 108 · antithrombin activity 75 · TT 37 · ACT 25
Study type	animal in vivo 1,430 · clinical 453 · in vitro 297 · ex vivo human plasma 205
Species	human 850 · monkey 818 · mouse 569 · rat 33 · pig 21 · minipig 17 · other 11 · not applicable 69
Effect direction	increase 702 · no change 573 · decrease 544 · not reported 449 · not applicable (pre-dose baseline) 120
Grade	ungraded 1,521 · 0 → 463 · 1 → 312 · 2 → 66 · 3 → 26 (155 grades flagged `within_reference_range_resolution`)
Axis	on-target only 1,576 · unintended only 289 · both 144 · neither 379

On negative controls. 573 rows carry a measured null (`effect_direction = no_change`), and they are the class most at risk of meaning "nobody looked". The extraction contract required that a null be written only where the source reports a measured null, with reporting silence recorded separately in `notes`; the verification pass tested this stratum specifically, because a prior review of the sibling kidney dataset found its negative class was substantially "nobody looked" rather than "looked and found nothing". **That defect does not repeat here**: four independent reviewers tried to break the null class and could not — the nulls are measured nulls, and unmeasured endpoints are typed `NOT_REPORTED` with notes saying so explicitly.

10. Quality control

Two committed scripts, both exiting non-zero on failure:

- `validate_dataset.py` — 45 structural checks (keys, referential integrity, vocabularies, grade reproducibility, sequence/modification consistency, roll-ups, and nine invariants added after verification). All pass. Defects caught during the build, and their fixes, are logged in `schema.md`.
- `verify_against_sources.py` — re-reads the committed documents and confirms every numeric readout appears in the source its row cites. **1,876 / 1,876 located.**

Structural QC proves internal consistency; source verification proves the numbers were not invented. Neither proves a number was read from the *right* cell, so a third pass did that by hand: 174 rows re-checked by reviewers instructed to refute them (117 confirmed, 50 corrected, 2 refuted, 5 unverifiable, no fabrication). Ten defect classes were found and corrected in the build; the residue is carried as open issues. Both the corrections and the residue are documented in `coagulopathy.md`.

11. Limitations

Stated in full in `README.md`. In brief: grades are provisional and mechanical; no clinical compound has a published sequence; PMO chemistry rests on regulatory silence rather than a measured null; prothrombotic rows come overwhelmingly from one compound family; the quantified PS class effect rests overwhelmingly on one study; figure-only values were not digitised.

12. Reproducibility

Deterministic and fully committed. From a clean checkout, with no network access: `build_dataset.py` → `validate_dataset.py` → `verify_against_sources.py`. The build reads the committed extraction records in `sources/extraction/` and writes `data/`; it uses only the Python standard library. Every number in this document and in `README.md` is recomputable from `data/`.

13. Intended use for predictive modelling

The four-table design exposes sequence, per-position chemistry and design predictors against graded, per-condition coagulation outcomes. The two axis flags are not optional metadata: any model trained across `on_target_effect = TRUE` rows without them will learn that anticoagulant drugs prolong clotting times, which is true, circular and useless for safety prediction. The intended target is the **unintended** class — the hybridization-independent, Cmax-driven prolongation associated with phosphorothioate content — with the on-target rows serving as a mechanistically-explained positive class and the 604 measured nulls as negatives.